COSET GRAPHS FOR MODULAR GROUP OVER CERTAIN GALOIS FIELDS

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Coset Graphs for Modular Group over Certain Galois Fields

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TABLE OF CONTENTS

1	Basic Concepts	1
2	Construction of $GF\ (2^5)$ and Coset graphs	3
3	Construction of $GF\ (2^4)$ and Coset graphs	29
4	Construction of $GF\ (3^3)$ and Coset graphs	35

Chapter 1

Basic Concepts

The initial chapter furnishes definitions and fundamental concepts in order to establish a framework for comprehending the principal findings.

1. *Finite Field*

In the pre-19th century era, there was a prevailing belief among algebraists that each finite field has prime order. Évariste Galois (1811–1832) developed Galois fields, denoted by $GF(p^n)$, which are finite fields of order p^n, where p is a prime number and n is a positive integer [1-2]. Each irreducible polynomial of degree n over Z_p produces a Galois field $GF(p^n)$, as demonstrated by Galois. Noting that there are multiple irreducible polynomials of degree n over Z_p for fixed n and p enables the development of multiple Galois fields $GF\ (p^n)$.

2. Hecke Group $H(3)$

The Hecke group $H(3)$ is an infinite group [3-4] of by linear transformations with two generators $x: z \longrightarrow \frac{-1}{z}$ and $y: z \longrightarrow \frac{(z-1)}{z}$ and two relations $x^2 = y^3 = 1$. Thus, the finite presentation of $H(3)$ is $\langle x, y: x^2 = y^3 = 1\rangle$.

3. Coset Graphs

These graphs are obtained by applying generators of Hecke group $H(3)$ on $GF(p^n) \cup \{\infty\}$. Both the generators $x: z \longrightarrow \frac{-1}{z}$ and $y: z \longrightarrow \frac{(z-1)}{z}$ of $H(3)$ are applied on each element of $GF(p^n)$. Consequently, we obtain permutation representations of x and y. Since x has order two, therefore, the permutation representation of x is the product of disjoint cycles of length 2. Similarly, the permutation y consists of length three disjoint cycles. In coset graph, each cycle (a, b, c) of y gives rise to a triangle abc and each cycle (r, s) of x represents a line joining r and s called x −edge. Note that, if r and s are in different cycles of y then (r, s) is a line joining vertices of two different triangles otherwise (r, s) is a loop, that is, a line joining two vertices of the same triangles. The elements of $GF(p^n) \cup \{\infty\}$ that are fixed points of x and y are presented by small circles. Thus, coset graph is a graph consisting of triangle such that each vertex of the triangle is connected with exactly one vertex of the triangle

through an x −edge. For more explanation about coset graphs of $H(3)$, we recommend to study [6-7].

Chapter Two

Construction of $GF\ (2^5)$ and Coset graphs

In this chapter, we will construct the all Galois fields of order 32. Then we compose coset graphs for Hecke group $H(3) = \langle x, y: x^2 = y^3 = 1\rangle$ over these fields.

To construct $GF\ (2^5)$, first we need to find all irreducible polynomial of degree 5 over Z_2. All such Polynomials are given below:

i. $f(x) = 1 + x^2 + x^5$

ii. $f(x) = 1 + x^3 + x^5$

iii. $f(x) = 1 + x + x^2 + x^3 + x^5$

iv. $f(x) = 1 + x + x^2 + x^4 + x^5$

v. $f(x) = 1 + x + x^3 + x^4 + x^5$

vi. $f(x) = 1 + x^2 + x^3 + x^4 + x^5$

1. *First Galois Field F_1 of Order 32*

The Table 1 displays all elements of $GF\ (2^5)$ evolved through $f(x) = 1 + x^3 + x^5$.

Table 1

Elements	**Polynomial Description**	**Digital Description**	**Elements**	**Polynomial Description**	**Digital Description**
x	x	01000	x^{17}	$x^3 + x^4$	00011
x^2	x^2	00100	x^{18}	$1 + x^3 + x^4$	10011
x^3	x^3	00010	x^{19}	$1 + x + x^3 + x^4$	11011
x^4	x^4	00001	x^{20}	$x + x^2 + x^3 + x^4 + 1$	11111
x^5	$1 + x^3$	10010	x^{21}	$1 + x + x^2 + x^4$	11101
x^6	$x + x^4$	01001	x^{22}	$1 + x^2 + x$	11100
x^7	$1 + x^2 + x^3$	10110	x^{23}	$x + x^2 + x^3$	01110
x^8	$x + x^4 + x^3$	01011	x^{24}	$x^2 + x^3 + x^4$	00111
x^9	$1 + x^2 + x^3 + x^4$	10111	x^{25}	$1 + x^4$	10001
x^{10}	$1 + x + x^4$	11001	x^{26}	$1 + x + x^3$	11010
x^{11}	$1 + x + x^2 + x^3$	11110	x^{27}	$x^2 + x + x^4$	01101
x^{12}	$x + x^2 + x^3 + x^4$	01111	x^{28}	$1 + x^2$	00100
x^{13}	$x^2 + 1 + x^4$	10101	x^{29}	$x + x^3$	01010
x^{14}	$1 + x$	11000	x^{30}	$x^2 + x^4$	00101
x^{15}	$x + x^2$	01100	x^{31}	1	10000

x^{16}	$x^2 + x^3$	00110			

In Table 2, 3 we apply the generators x and y of $H(3)$ on all elements of F_1.

Table 2

Transformation $x(z)$	**Action**	**Permutation**
$x(x)$	$-\frac{1}{x} = \frac{1}{x} = x^{30}$	(x, x^{30})
$x(x^2)$	$-\frac{1}{x^2} = \frac{1}{x^2} = x^{29}$	(x^2, x^{29})
$x(x^3)$	$\frac{-1}{x^3} = \frac{1}{x^3} = x^{28}$	(x^3, x^{28})
$x(x^4)$	$-\frac{1}{x^4} = \frac{1}{x^4} = x^{27}$	(x^4, x^{27})
$x(x^5)$	$-\frac{1}{x^5} = \frac{1}{x^5} = x^{26}$	(x^5, x^{26})
$x(x^6)$	$-\frac{1}{x^6} = \frac{1}{x^6} = x^{25}$	(x^6, x^{25})
$x(x^7)$	$-\frac{1}{x^7} = \frac{1}{x^7} = x^{24}$	(x^7, x^{24})

$x(x^8)$	$\frac{-1}{x^8} = \frac{1}{x^8} = x^{23}$	(x^8, x^{23})
$x(x^9)$	$\frac{-1}{x^9} = \frac{1}{x^9} = x^{22}$	(x^9, x^{22})
$x(x^{10})$	$\frac{-1}{x^{10}} = \frac{1}{x^{10}} = x^{21}$	(x^{10}, x^{21})
$x(x^{11})$	$\frac{-1}{x^{11}} = \frac{1}{x^{11}} = x^{20}$	(x^{11}, x^{20})
$x(x^{12})$	$\frac{-1}{x^{12}} = \frac{1}{x^{12}} = x^{19}$	(x^{12}, x^{19})
$x(x^{13})$	$\frac{-1}{x^{13}} = \frac{1}{x^{13}} = x^{18}$	(x^{13}, x^{18})
$x(x^{14})$	$\frac{-1}{x^{14}} = \frac{1}{x^{14}} = x^{17}$	(x^{14}, x^{17})
$x(x^{15})$	$\frac{-1}{x^{15}} = \frac{1}{x^{15}} = x^{16}$	(x^{15}, x^{16})
$x(x^{31})$	$\frac{-1}{x^{31}} = \frac{1}{x^{31}} = x^{31}$	(x^{31})
$x(0)$	$\frac{1}{0} = \infty$	$(0, \infty)$

Table 3

Transformation $\boldsymbol{y(z)}$	**Action**	**Permutations**
$y(x)$	$\frac{x+1}{x} = x^{13}$ $\frac{1+x^{13}}{x^{13}} = x^{17}$	(x, x^{13}, x^{17})
$y(x^2)$	$\frac{x^2+1}{x^2} = x^{26}$ $\frac{x^{26}+1}{x^{26}} = x^3$	(x^2, x^{26}, x^3)
$y(x^4)$	$\frac{x^4+1}{x^4} = x^{21}$ $\frac{x^{21}+1}{x^{21}} = x^6$	(x^4, x^{21}, x^6)
$y(x^5)$	$\frac{x^5+1}{x^5} = x^{29}$ $\frac{x^{29}+1}{x^{29}} = x^{28}$	(x^5, x^{29}, x^{28})
$y(x^7)$	$\frac{x^7+1}{x^7} = x^9$ $\frac{x^9+1}{x^9} = x^{15}$	(x^7, x^9, x^{15})
$y(x^8)$	$\frac{x^8+1}{x^8} = x^{11}$ $\frac{x^{11}+1}{x^{11}} = x^{12}$	(x^8, x^{11}, x^{12})
$y(x^{10})$	$(x^{10}+1)\frac{1}{x^{10}} = x^{27}$ $\frac{x^{27}+1}{x^{27}} = x^{25}$	(x^{10}, x^{27}, x^{25})
$y(x^{14})$	$\frac{x^{14}+1}{x^{14}} = x^{18}$ $\frac{x^{18}+1}{x^{18}} = x^{30}$	(x^{14}, x^{18}, x^{30})

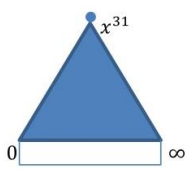
$y(x^{16})$	$\frac{x^{16}+1}{x^{16}} = x^{22}$ $\frac{x^{22}+1}{x^{22}} = x^{24}$	(x^{16}, x^{22}, x^{24})
$y(x^{19})$	$\frac{x^{19}+1}{x^{19}} = x^{20}$ $\frac{x^{20}+1}{x^{20}} = x^{23}$	(x^{19}, x^{20}, x^{23})
$y(x^{31})$	$\frac{x^{31}+1}{x^{31}} = x^{0}$ $\frac{0+1}{0} = \infty$	$(x^{31}, 0, \infty)$

Using Table 2,3 we have the following permutations of x and y

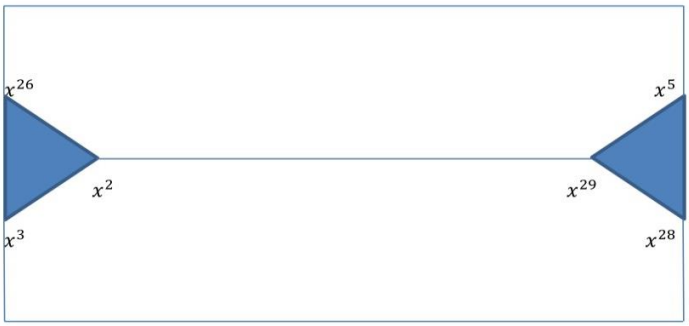
$$\boldsymbol{x} = (x, x^{30})(x^2, x^{29})(x^3, x^{28})(x^4, x^{27})(x^5, x^{26})(x^6, x^{25})(x^7, x^{24})(x^8, x^{23})$$
$$(x^9, x^{22})(x^{10}, x^{21})(x^{11}, x^{20})(x^{12}, x^{19})(x^{13}, x^{18})(x^{14}, x^{17})(x^{15}, x^{16})(x^{31})(0, \infty)$$

$$\boldsymbol{y} = (x, x^{13}, x^{17})(x^2, x^{26}, x^3)(x^4, x^{21}, x^6)(x^5, x^{29}, x^{28})(x^7, x^9, x^{15})$$
$$(x^8, x^{11}, x^{12})(x^{10}, x^{27}, x^{25})(x^{14}, x^{18}, x^{30})(x^{16}, x^{22}, x^{24})(x^{19}, x^{20}, x^{23})$$
$$(x^{31}, 0, \infty)$$

These permutations combine to produce the following coset graph:

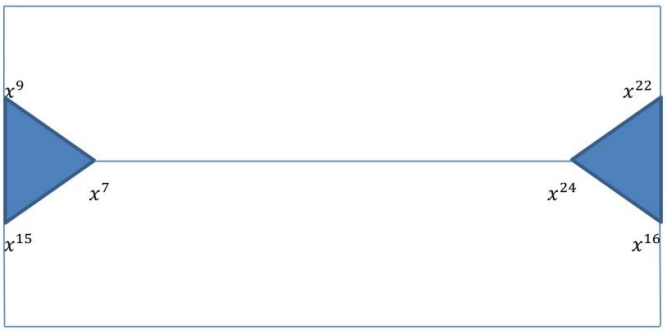


x^{13}
x^{18}
x^{1}
x^{30}
x^{17}
x^{14}


x^{26}
x^{5}
x^{2}
x^{29}
x^{3}
x^{28}

x^{21}
x^{10}
x^{4}
x^{27}
x^{6}
x^{25}

x^{11}
x^{20}
x^{8}
x^{23}
x^{12}
x^{19}



2 *Second Galois Field F_2 of Order 32*

The Table 4 displays all elements of $GF\ (2^5)$ evolved through $f(x) = 1 + x + x^2 + x^3 + x^5$

Table 4

Elements	**Polynomial Description**	**Digital Description**	**Elements**	**Polynomial Description**	**Digital Description**
x	x	01000	x^{17}	$1 + x^4$	10001
x^2	x^2	00100	x^{18}	$1 + x^2 + x^3$	10110
x^3	x^3	00010	x^{19}	$x + x^4 + x^3$	01011
x^4	x^4	00001	x^{20}	$x + x^3 + 1 + x^4$	11011
x^5	$1 + x^2 + x + x^3$	11110	x^{21}	$x^3 + x^4 + 1$	10011
x^6	$x + x^3 + x^2 + x^4$	01111	x^{22}	$1 + x^2 + x^4 + x^3$	10111
x^7	$1 + x + x^4$	11001	x^{23}	$1 + x^2 + x^4$	10101
x^8	$1 + x^3$	10010	x^{24}	$1 + x^2$	10100
x^9	$x + x^4$	01001	x^{25}	$x + x^3$	01010
x^{10}	$1 + x + x^3$	11010	x^{26}	$x^2 + x^4$	00101
x^{11}	$x + x^4 + x^2$	01101	x^{27}	$1 + x + x^2$	11100
x^{12}	$x + 1$	11000	x^{28}	$x + x^2 + x^3$	01110
x^{13}	$x^2 + x$	01100	x^{29}	$x^2 + x^3 + x^4$	00111
x^{14}	$x^2 + x^3$	00110	x^{30}	$1 + x + x2 + x4$	11101
x^{15}	$x^3 + x^4$	00011	x^{31}	1	10000
x^{16}	$1 + x + x^2 + x^3 + x^4$	11111			

In Table 5, 6 we apply the generators x and y of $H(3)$ on all elements of F_2.

Table 5

Transformation $x(z)$	**Action**	**Permutation**

$x(x)$	$-\frac{1}{x}=\frac{1}{x}=x^{30}$	(x,x^{30})
$x(x^2)$	$-\frac{1}{x^2}=\frac{1}{x^2}=x^{29}$	(x^2,x^{29})
$x(x^3)$	$\frac{-1}{x^3}=\frac{1}{x^3}=x^{28}$	(x^3,x^{28})
$x(x^4)$	$-\frac{1}{x^4}=\frac{1}{x^4}=x^{27}$	(x^4,x^{27})
$x(x^5)$	$-\frac{1}{x^5}=\frac{1}{x^5}=x^{26}$	(x^5,x^{26})
$x(x^6)$	$-\frac{1}{x^6}=\frac{1}{x^6}=x^{25}$	(x^6,x^{25})
$x(x^7)$	$-\frac{1}{x^7}=\frac{1}{x^7}=x^{24}$	(x^7,x^{24})
$x(x^8)$	$\frac{-1}{x^8}=\frac{1}{x^8}=x^{23}$	(x^8,x^{23})
$x(x^9)$	$\frac{-1}{x^9}=\frac{1}{x^9}=x^{22}$	(x^9,x^{22})

$x(x^{10})$	$\frac{-1}{x^{10}} = \frac{1}{x^{10}} = x^{21}$	(x^{10}, x^{21})
$x(x^{11})$	$\frac{-1}{x^{11}} = \frac{1}{x^{11}} = x^{20}$	(x^{11}, x^{20})
$x(x^{12})$	$\frac{-1}{x^{12}} = \frac{1}{x^{12}} = x^{19}$	(x^{12}, x^{19})
$x(x^{13})$	$\frac{-1}{x^{13}} = \frac{1}{x^{13}} = x^{18}$	(x^{13}, x^{18})
$x(x^{14})$	$\frac{-1}{x^{14}} = \frac{1}{x^{14}} = x^{17}$	(x^{14}, x^{17})
$x(x^{15})$	$\frac{-1}{x^{15}} = \frac{1}{x^{15}} = x^{16}$	(x^{15}, x^{16})
$x(x^{31})$	$\frac{-1}{x^{31}} = \frac{1}{x^{31}} = x^{31}$	(x^{31})
$x(0)$	$\frac{1}{0} = \infty$	$(0, \infty)$

Transformation $y(z)$	**Action**	**Permutation**
$y(x)$	$\frac{x+1}{x} = x^{11}$ $1 + x^{20} = x^{19}$	(x, x^{11}, x^{19})
$y(x^2)$	$\frac{x^2+1}{x^2} = x^{22}$ $\frac{x^{22}+1}{x^{22}} = x^7$	(x^2, x^{22}, x^7)
$y(x^3)$	$\frac{x^3+1}{x^3} = x^5$ $\frac{x^5+1}{x^5} = x^{23}$	(x^3, x^5, x^{23})
$y(x^4)$	$\frac{x^4+1}{x^4} = x^{13}$ $\frac{x^{13}+1}{x^{13}} = x^{14}$	(x^4, x^{13}, x^{14})
$y(x^6)$	$\frac{x^6+1}{x^6} = x^{10}$ $\frac{x^{10}+1}{x^{10}} = x^{15}$	(x^6, x^{10}, x^{15})
$y(x^8)$	$\frac{x^8+1}{x^8} = x^{26}$ $\frac{x^{26}+1}{x^{26}} = x^{28}$	(x^8, x^{26}, x^{28})
$y(x^9)$	$\frac{x^9+1}{x^9} = x^{29}$ $\frac{x^{29}+1}{x^{29}} = x^{24}$	(x^9, x^{29}, x^{24})
$y(x^{16})$	$\frac{x^{16}+1}{x^{16}} = x^{21}$ $\frac{x^{21}+1}{x^{21}} = x^{25}$	(x^{16}, x^{21}, x^{25})
$y(x^{17})$	$\frac{x^{17}+1}{x^{17}} = x^{18}$ $\frac{x^{18}+1}{x^{18}} = x^{27}$	(x^{17}, x^{18}, x^{27})
	$\frac{x^{31}+1}{x^{31}} = x^0$	

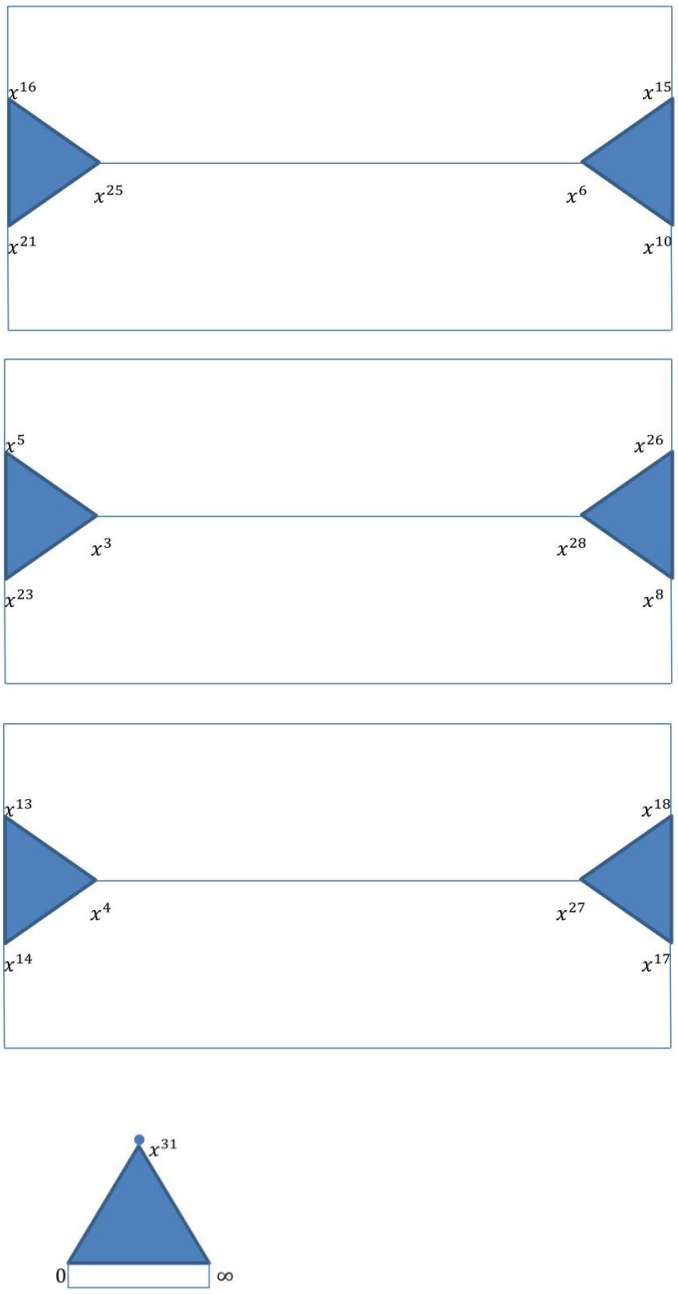
$y(x^{31})$	$\frac{0+1}{0} = \infty$	$(x^{31}, 0, \infty)$
$y(x^{12})$	$\frac{x^{12}+1}{x^{12}} = x^{20}$ $\frac{x^{20}+1}{x^{20}} = x^{30}$	(x^{12}, x^{20}, x^{30})

Using Table 5, 6 we have the following permutations x of y

$$x = (x, x^{30})(x^2, x^{29})(x^3, x^{28})(x^4, x^{27})(x^5, x^{26})(x^6, x^{25})(x^7, x^{24})(x^8, x^{23})$$
$$(x^9, x^{22})(x^{10}, x^{21})(x^{11}, x^{20})(x^{12}, x^{19})(x^{13}, x^{18})(x^{14}, x^{17})(x^{15}, x^{16})(x^{31})(0, \infty)$$

$$y = (x, x^{11}, x^{19})(x^2, x^{22}, x^7)(x^3, x^5, x^{23})(x^4, x^{13}, x^{14})(x^6, x^{10}, x^{15})$$
$$(x^8, x^{26}, x^{28})(x^9, x^{29}, x^{24})(x^{16}, x^{21}, x^{25})(x^{17}, x^{18}, x^{27})(x^{31,0}, \infty)(x^{12}, x^{20}, x^{30})$$

These permutations combine to produce the following coset graph:



3. *Third Galois Field F_3 of Order 32*

The Table 7 displays all elements of $GF\ (2^5)$ evolved through $f(x) = 1 + x^2 + x^5$

Table 7

Elements	**Polynomial Description**	**Digital Description**	**Elements**	**Polynomial Description**	**Digital Description**
x	x	01000	x^{17}	$1+x+x^4$	11001
x^2	x^2	00100	x^{18}	$1+x$	11000
x^3	x^3	00010	x^{19}	$x+x^2$	01100
x^4	x^4	00001	x^{20}	x^2+x^3	00110
x^5	$1+x^2$	10100	x^{21}	x^3+x^4	00011
x^6	$x+x^3$	01010	x^{22}	$1+x^2+x^4$	10101
x^7	x^2+x^4	01010	x^{23}	$1+x+x^2+x^3$	11110
x^8	$1+x^2+x^3$	10001	x^{24}	$x+x^2+x^3+x^4$	01111
x^9	$x+x^3+x^4$	01011	x^{25}	$1+x^3+x^4$	10011
x^{10}	$1+x^4$	10001	x^{26}	$1+x+x^2+x^4$	11101
x^{11}	$1+x+x^2$	11100	x^{27}	$1+x+x^3$	11010
x^{12}	$x+x^2+x^3$	01110	x^{28}	$x+x^2+x^4$	01101
x^{13}	$x^2+x^3+x^4$	00111	x^{29}	$1+x^3$	10010
x^{14}	$1+x^2+x^3+x^4$	10111	x^{30}	$x+x^4$	01001
x^{15}	$1+x+x^2+x^3+x^4$	11111	x^{31}	1	10000
x^{16}	$1+x+x^3+x^4$	11011			

we apply the generators x and y of $H(3)$ on all elements of F_3

Table 8

Transformation $x(z)$	**Action**	**Permutation**
$x(x)$	$-\frac{1}{x}=\frac{1}{x}=x^{30}$	(x,x^{30})
$x(x^2)$	$-\frac{1}{x^2}=\frac{1}{x^2}=x^{29}$	(x^2,x^{29})
$x(x^3)$	$\frac{-1}{x^3}=\frac{1}{x^3}=x^{28}$	(x^3,x^{28})

$x(x^4)$	$-\frac{1}{x^4}=\frac{1}{x^4}=x^{27}$	(x^4,x^{27})
$x(x^5)$	$-\frac{1}{x^5}=\frac{1}{x^5}=x^{26}$	(x^5,x^{26})
$x(x^6)$	$-\frac{1}{x^6}=\frac{1}{x^6}=x^{25}$	(x^6,x^{25})
$x(x^7)$	$-\frac{1}{x^7}=\frac{1}{x^7}=x^{24}$	(x^7,x^{24})
$x(x^8)$	$\frac{-1}{x^8}=\frac{1}{x^8}=x^{23}$	(x^8,x^{23})
$x(x^9)$	$\frac{-1}{x^9}=\frac{1}{x^9}=x^{22}$	(x^9,x^{22})
$x(x^{10})$	$\frac{-1}{x^{10}}=\frac{1}{x^{10}}=x^{21}$	(x^{10},x^{21})
$x(x^{11})$	$\frac{-1}{x^{11}}=\frac{1}{x^{11}}=x^{20}$	(x^{11},x^{20})
$x(x^{12})$	$\frac{-1}{x^{12}}=\frac{1}{x^{12}}=x^{19}$	(x^{12},x^{19})
$x(x^{13})$		(x^{13},x^{18})

	$\frac{-1}{x^{13}} = \frac{1}{x^{13}} = x^{18}$	
$x(x^{14})$	$\frac{-1}{x^{14}} = \frac{1}{x^{14}} = x^{17}$	(x^{14}, x^{17})
$x(x^{15})$	$\frac{-1}{x^{15}} = \frac{1}{x^{15}} = x^{16}$	(x^{15}, x^{16})
$x(x^{31})$	$\frac{-1}{x^{31}} = \frac{1}{x^{31}} = x^{31}$	(x^{31})
$x(0)$	$\frac{-1}{0} = \frac{1}{0} = \infty$	$(0, \infty)$

Table 9

Transformation $y(z)$	**Action**	**Permutation**
$y(x)$	$\frac{x+1}{x} = x^{17}$ $\frac{1+x^{17}}{x^{17}} = x^{13}$	$(x,, x^{17}, x^{13})$
$y(x^2)$	$\frac{x^2+1}{x^2} = x^3$ $\frac{x^3+1}{x^3} = x^{26}$	$(x^2,, x^3, x^{26})$
$y(x^4)$	$\frac{x^4+1}{x^4} = x^6$ $\frac{x^6+1}{x^6} = x^{21}$	(x^3, x^6, x^{21})

$y(x^5)$	$\frac{x^5+1}{x^5} = x^{28}$ $\frac{x^{28}+1}{x^{28}} = x^{29}$	(x^5, x^{28}, x^{29})
$y(x^7)$	$\frac{x^7+1}{x^7} = x^{15}$ $\frac{x^{15}+1}{x^{15}} = x^9$	(x^7, x^{15}, x^9)
$y(x^8)$	$\frac{x^8+1}{x^8} = x^{12}$ $\frac{x^{12}+1}{x^{12}} = x^{11}$	(x^8, x^{12}, x^{11})
$y(x^{10})$	$(x^{10}+1)\frac{1}{x^{10}}$ $= x^{25}$ $\frac{x^{25}+1}{x^{25}} = x^{27}$	$(x^{10}, x^{25}x^{27})$
$y(x^{14})$	$\frac{x^{14}+1}{x^{14}} = x^{30}$ $\frac{x^{30}+1}{x^{30}} = x^{18}$	(x^{14}, x^{30}, x^{18})
$y(x^{16})$	$\frac{x^{16}+1}{x^{16}} = x^{24}$ $\frac{x^{24}+1}{x^{24}} = x^{22}$	(x^{16}, x^{24}, x^{22})
$y(x^{19})$	$\frac{x^{19}+1}{x^{19}} = x^{23}$ $\frac{x^{23}+1}{x^{23}} = x^{20}$	(x^{19}, x^{23}, x^{20})

$y(x^{31})$	$\frac{x^{31}+1}{x^{31}} = x^0$ $\frac{0+1}{0} = \infty$	$(x^{31}, 0, \infty)$

By using tables 8, 9 we have following permutations of x and y

$$x = (x, x^{30})(x^2, x^{29})(x^3, x^{28})(x^4, x^{27})(x^5, x^{26})(x^6, x^{25})(x^7, x^{24})(x^8, x^{23})$$

$$(x^9, x^{22})(x^{10}, x^{21})(x^{11}, x^{20})(x^{12}, x^{19})(x^{13}, x^{18})(x^{14}, x^{17})(x^{15}, x^{16})(x^{31})(0, \infty)$$

$$y = (x,, x^{17}, x^{13})(x^2,, x^3, x^{26})(x^3, x^6, x^{21})(x^5, x^{28}, x^{29})(x^7, x^{15}, x^9)(x^8, x^{12}, x^{11})$$
$$(x^{10}, x^{25} x^{27})(x^{14}, x^{30}, x^{18})(x^{16}, x^{24}, x^{22})(x^{19}, x^{23}, x^{20})(x^{31}, 0, \infty)$$

Now, by using the permutation of x and y one can easily make corresponding coset graphs

4. *Fourth Galois Field F_4 of Order 32*

The Table 10 displays all elements of $GF\ (2^5)$ evolved through $f(x) = 1 + x^2 + x^4 + x^5 + x$

Table 10

Elements	**Polynomial Description**	**Digital Description**	**Elements**	**Polynomial Description**	**Digital Description**
x	x	01000	x^{17}	$1 + x^2 + x^3$	10110
x^2	x^2	00100	x^{18}	$x + x^3 + x^4$	01011
x^3	x^3	00010	x^{19}	$x + 1$	11000
x^4	x^4	00001	x^{20}	$x^2 + x$	01100
x^5	$1 + x^2 + x + x^4$	11101	x^{21}	$x^3 + x^2$	00110
x^6	$1 + x^4 + x^3$	10011	x^{22}	$x^3 + x^4$	00011
x^7	$1 + x^2$	10100	x^{23}	$1 + x + x^2 + x^3$	11100
x^8	$x + x^3$	01010	x^{24}	$1 + x + x^2$	01110
x^9	$x^2 + x^4$	00101	x^{25}	$x + x^3 + x^2$	00111
x^{10}	$1 + x + x^2 + x^3 + x^4$	11111	x^{26}	$1 + x + x^2 + x^3$	11110
x^{11}	$1 + x^3$	10010	x^{27}	$x^4 + x + x^3 + x^2$	01111
x^{12}	$x + x^4$	01001	x^{28}	$1 + x + x^3$	11010
x^{13}	$1 + x + x^4$	11001	x^{29}	$x + x^2 + x^4$	01101
x^{14}	$1 + x^4$	10001	x^{30}	$1 + x + x^3 + x^4$	11011
x^{15}	$1 + x^2 + x^4$	10101	x^{31}	1	10000

x^{16}	$1+x^2+x^3+x^4$	10111			

we apply the generators x and y of $H(3)$ on all elements of F_4

Table 11

Transformation $x(z)$	**Action**	**Permutation**
$x(x)$	$-\frac{1}{x}=\frac{1}{x}=x^{30}$	(x,x^{30})
$x(x^2)$	$-\frac{1}{x^2}=\frac{1}{x^2}=x^{29}$	(x^2,x^{29})
$x(x^3)$	$\frac{-1}{x^3}=\frac{1}{x^3}=x^{28}$	(x^3,x^{28})
$x(x^4)$	$-\frac{1}{x^4}=\frac{1}{x^4}=x^{27}$	(x^4,x^{27})
$x(x^5)$	$-\frac{1}{x^5}=\frac{1}{x^5}=x^{26}$	(x^5,x^{26})
$x(x^6)$	$-\frac{1}{x^6}=\frac{1}{x^6}=x^{25}$	(x^6,x^{25})

$x(x^7)$	$-\frac{1}{x^7}=\frac{1}{x^7}=x^{24}$	(x^7,x^{24})
$x(x^8)$	$\frac{-1}{x^8}=\frac{1}{x^8}=x^{23}$	(x^8,x^{23})
$x(x^9)$	$\frac{-1}{x^9}=\frac{1}{x^9}=x^{22}$	(x^9,x^{22})
$x(x^{10})$	$\frac{-1}{x^{10}}=\frac{1}{x^{10}}=x^{21}$	(x^{10},x^{21})
$x(x^{11})$	$\frac{-1}{x^{11}}=\frac{1}{x^{11}}=x^{20}$	(x^{11},x^{20})
$x(x^{12})$	$\frac{-1}{x^{12}}=\frac{1}{x^{12}}=x^{19}$	(x^{12},x^{19})
$x(x^{13})$	$\frac{-1}{x^{13}}=\frac{1}{x^{13}}=x^{18}$	(x^{13},x^{18})
$x(x^{14})$	$\frac{-1}{x^{14}}=\frac{1}{x^{14}}=x^{17}$	(x^{14},x^{17})
$x(x^{15})$	$\frac{-1}{x^{15}}=\frac{1}{x^{15}}=x^{16}$	(x^{15},x^{16})
$x(x^{31})$	$\frac{-1}{x^{31}}=\frac{1}{x^{31}}=x^{31}$	(x^{31})
$x(0)$	$\frac{-1}{0}=\frac{1}{0}=\infty$	$(0,\infty)$

Table 12

Transformatio	**Action**	**Permutation**

n $\boldsymbol{y(z)}$		
$y(x)$	$\frac{x+1}{x} = x^{18}$ $1 + x^{11} = x^{12}$	$(x, x^{18}x^{12})$
$y(x^2)$	$\frac{x^2+1}{x^2} = x^5$ $\frac{x^5+1}{x^5} = x^{24}$	(x^2, x^5, x^{24})
$y(x^3)$	$\frac{x^3+1}{x^3} = x^8$ $\frac{x^8+1}{x^8} = x^{20}$	(x^3, x^8, x^{20})
$y(x^4)$	$\frac{x^4+1}{x^4} = x^{10}$ $\frac{x^{10}+1}{x^{10}} = x^{17}$	(x^4, x^{10}, x^{17})
$y(x^6)$	$\frac{x^6+1}{x^6} = x^{16}$ $\frac{x^{16}+1}{x^{16}} = x^{19}$	(x^6, x^{16}, x^{19})
$y(x^7)$	$\frac{x^7+1}{x^7} = x^{26}$ $\frac{x^{26}+1}{x^{26}} = x^{29}$	$(x^7, x^{26}x^{29})$
$y(x^{11})$	$(x^{11}+1)\frac{1}{x^{11}}$ $= x^{23}$ $\frac{x^{23}+1}{x^{23}} = x^{28}$	(x^{11}, x^{23}, x^{28})

$y(x^{13})$	$\frac{x^{13}+1}{x^{13}} = x^{30}$ $\frac{x^{30}+1}{x^{30}} = x^{19}$	(x^{13}, x^{30}, x^{19})
$y(x^{14})$	$\frac{x^{14}+1}{x^{14}} = x^{21}$ $\frac{x^{21}+1}{x^{21}} = x^{27}$	(x^{14}, x^{21}, x^{27})
$y(x^{15})$	$\frac{x^{15}+1}{x^{15}} = x^{25}$ $\frac{x^{25}+1}{x^{25}} = x^{22}$	(x^{15}, x^{25}, x^{22})
$y(x^{31})$	$\frac{x^{31}+1}{x^{31}} = 0$ $\frac{0+1}{0} = \infty$	$(x^{31}, 0, \infty)$

By using table 11, 12 we have following permutation of x and y

$$x = (x, x^{30})(x^2, x^{29})(x^3, x^{28})(x^4, x^{27})(x^5, x^{26})(x^6, x^{25})(x^7, x^{24})(x^8, x^{23})$$
$$(x^9, x^{22})(x^{10}, x^{21})(x^{11}, x^{20})(x^{12}, x^{19})(x^{13}, x^{18})(x^{14}, x^{17})(x^{15}, x^{16})(x^{31})(0, \infty)$$

$$y = (x, x^{18}x^{12})(x^2, x^5, x^{24})(x^3, x^8, x^{20})(x^4, x^{10}, x^{17})(x^6, x^{16}, x^{19})$$
$$(x^7, x^{26}x^{29})(x^{11}, x^{23}, x^{28})(x^{13}, x^{30}, x^{19})(x^{14}, x^{21}, x^{27})(x^{15}, x^{25}, x^{22})(x^{31}, 0, \infty)$$

Now, by using the permutation of x and y one can easily make corresponding coset graphs

5. *Fifth Galois Field F_5 of Order 32*

The Table 13 displays all elements of $GF\ (2^5)$ evolved through
$f(x) = 1 + x^2 + x^3 + x^4 + x^5$

Table 13

Elements	**Polynomial Description**	**Digital Description**	**Elements**	**Polynomial Description**	**Digital Description**
x	x	01000	x^{17}	$x + x^2 + x^4$	01101

x^2	x^2	00100	x^{18}	$1+x^4$	10001
x^3	x^3	00010	x^{19}	$1+x^2+x+x^3+x^4$	11111
x^4	x^4	00001	x^{20}	$x+1$	11000
x^5	$1+x^2+x^4+x^3$	10111	x^{21}	x^2+x	01100
x^6	x^2+x+1	11100	x^{22}	x^3+x^4	00110
x^7	$x+x^2+x^3$	01110	x^{23}	x^4+x^3	00011
x^8	$x^2+x^3+x^4$	00111	x^{24}	$1+x^2+x^3$	10110
x^9	$1+x^2$	10100	x^{25}	$x+x^3+x^4$	01011
x^{10}	$x+x^3$	01010	x^{26}	$1+x^3$	10010
x^{11}	x^4+x^2	00101	x^{27}	$x+x^4$	01001
x^{12}	$1+x^2+x^4$	10101	x^{28}	$1+x^3+x^4$	10011
x^{13}	$1+x^2+x+x^4$	11101	x^{29}	$1+x^2+x^3+x$	11110
x^{14}	$1+x+x^4$	11001	x^{30}	$x+x^4+x^2+x^3$	01111
x^{15}	$1+x+x^3+x^4$	11011	x^{31}	1	10000
x^{16}	$1+x+x^3$	11010			

we apply the generators x and y of $H(3)$ on all elements of F_5

Table 14

Transformation $x(z)$	**Action**	**Permutation**
$x(x)$	$-\frac{1}{x}=\frac{1}{x}=x^{30}$	(x,x^{30})
$x(x^2)$	$-\frac{1}{x^2}=\frac{1}{x^2}=x^{29}$	(x^2,x^{29})
$x(x^3)$	$\frac{-1}{x^3}=\frac{1}{x^3}=x^{28}$	(x^3,x^{28})

$x(x^4)$	$-\frac{1}{x^4}=\frac{1}{x^4}=x^{27}$	(x^4,x^{27})
$x(x^5)$	$-\frac{1}{x^5}=\frac{1}{x^5}=x^{26}$	(x^5,x^{26})
$x(x^6)$	$-\frac{1}{x^6}=\frac{1}{x^6}=x^{25}$	(x^6,x^{25})
$x(x^7)$	$-\frac{1}{x^7}=\frac{1}{x^7}=x^{24}$	(x^7,x^{24})
$x(x^8)$	$\frac{-1}{x^8}=\frac{1}{x^8}=x^{23}$	(x^8,x^{23})
$x(x^9)$	$\frac{-1}{x^9}=\frac{1}{x^9}=x^{22}$	(x^9,x^{22})
$x(x^{10})$	$\frac{-1}{x^{10}}=\frac{1}{x^{10}}=x^{21}$	(x^{10},x^{21})
$x(x^{11})$	$\frac{-1}{x^{11}}=\frac{1}{x^{11}}=x^{20}$	(x^{11},x^{20})
$x(x^{12})$	$\frac{-1}{x^{12}}=\frac{1}{x^{12}}=x^{19}$	(x^{12},x^{19})
$x(x^{13})$		(x^{13},x^{18})

	$\frac{-1}{x^{13}} = \frac{1}{x^{13}} = x^{18}$	
$x(x^{14})$	$\frac{-1}{x^{14}} = \frac{1}{x^{14}} = x^{17}$	(x^{14}, x^{17})
$x(x^{15})$	$\frac{-1}{x^{15}} = \frac{1}{x^{15}} = x^{16}$	(x^{15}, x^{16})
$x(x^{31})$	$\frac{1}{x^{31}}$	(x^{31})
$x(0)$	$\frac{1}{0} = \infty$	$(0, \infty)$

Table 15

Transformation $\boldsymbol{y(z)}$	**Action**	**Permutation**
$y(x)$	$\frac{x+1}{x} = x^{19}$ $1 + x^{12} = x^{11}$	$(x, x^{19}x^{11})$

$y(x^2)$	$\frac{x^2+1}{x^2} = x^7$ $\frac{x^7+1}{x^7} = x^{22}$	(x^2, x^7, x^{22})
$y(x^3)$	$\frac{x^3+1}{x^3} = x^{23}$ $\frac{x^{23}+1}{x^{23}} = x^5$	(x^3, x^{23}, x^5)
$y(x^4)$	$\frac{x^4+1}{x^4} = x^{14}$ $\frac{x^{14}+1}{x^{14}} = x^{13}$	(x^4, x^{14}, x^{13})
$y(x^6)$	$\frac{x^6+1}{x^6} = x^{15}$ $\frac{x^{15}+1}{x^{15}} = x^{10}$	(x^6, x^{15}, x^{10})
$y(x^8)$	$\frac{x^8+1}{x^8} = x^{28}$ $\frac{x^{28}+1}{x^{28}} = x^{26}$	$(x^8, x^{28}x^{26})$
$y(x^9)$	$(x^9+1)\frac{1}{x^9}$ $= x^{24}$ $\frac{x^{24}+1}{x^{24}} = x^{29}$	(x^9, x^{24}, x^{29})
$y(x^{12})$	$\frac{x^{12}+1}{x^{12}} = x^{30}$ $\frac{x^{30}+1}{x^{30}} = x^{20}$	(x^{12}, x^{30}, x^{20})

$y(x^{16})$	$\frac{x^{16}+1}{x^{16}} = x^{25}$ $\frac{x^{25}+1}{x^{25}} = x^{21}$	(x^{16}, x^{25}, x^{21})
$y(x^{17})$	$\frac{x^{17}+1}{x^{17}} = x^{27}$ $\frac{x^{27}+1}{x^{27}} = x^{18}$	(x^{17}, x^{27}, x^{18})
$y(x^{31})$	$\frac{x^{31}+1}{x^{31}} = 0$ $\frac{0+1}{0} = \infty$	$(x^{31}, 0, \infty)$

By using table 14, 15 we have following permutation of x and y

$$x = (x, x^{30})(x^2, x^{29})(x^3, x^{28})(x^4, x^{27})(x^5, x^{26})(x^6, x^{25})(x^7, x^{24})(x^8, x^{23})$$
$$(x^9, x^{22})(x^{10}, x^{21})(x^{11}, x^{20})(x^{12}, x^{19})(x^{13}, x^{18})(x^{14}, x^{17})(x^{15}, x^{16})(x^{31})(0, \infty)$$

$$y = (x, x^{19}x^{11})(x^2, x^7, x^{22})(x^3, x^{23}, x^5)(x^4, x^{14}, x^{13})(x^6, x^{15}, x^{10})(x^8, x^{28}x^{26})$$
$$(x^9, x^{24}, x^{29})(x^{12}, x^{30}, x^{20})(x^{16}, x^{25}, x^{21})(x^{17}, x^{27}, x^{18})(x^{31}, 0, \infty)$$

Now, by using the permutation of x and y one can easily make corresponding coset graphs

Chapter Three

Construction of $GF\ (2^4)$ and Coset graphs

Galois Field of Order 16

In this chapter, we will construct the all Galois fields of order 16. Then we compose coset graphs for Hecke group $H(3) = \langle x, y: x^2 = y^3 = 1\rangle$ over these fields.

To construct $GF\ (2^4)$, first we need to find all irreducible polynomial of degree 4 over Z_2. All such Polynomials are given below:

i. $f(x) = 1 + x + x^4$
ii. $f(x) = 1 + x^3 + x^4$

1. First Galois Field F_1 of Order 16

The Table 1 displays all elements of $GF\ (2^4)$ evolved through $f(x) = 1 + x + x^4$

Table 1

Elements	**Polynomial Description**	**Digital description**	**Elements**	**Polynomial Description**	**Digital Description**
x	x	0100	x^9	$x + x^3$	0101
x^2	x^2	0010	x^{10}	$1 + x + x^2$	1110
x^3	x^3	0001	x^{11}	$x + x^2 + x^3$	0111
x^4	$1 + x$	1100	x^{12}	$x + x^2 + x^3 + 1$	1111
x^5	$x + x^2$	0110	x^{13}	$x^2 + x^3 + 1$	1011
x^6	$x^2 + x^3$	0011	x^{14}	$x^3 + 1$	1001
x^7	$1 + x + x^3$	1101	x^{15}	1	1000
x^8	$x^2 + 1$	1010			

In Table 2, 3 we apply the generators x and y of $H(3)$ on all elements of F_1.

Table 2

Transformation $x(z)$	**Action**	**Permutation**

$x(x)$	$\frac{-1}{x} = \frac{1}{x} = x^{14}$	(x, x^{14})
$x(x^2)$	$\frac{-1}{x^2} = \frac{1}{x^2} = x^{13}$	(x^2, x^{13})
$x(x^3)$	$\frac{-1}{x^3} = \frac{1}{x^3} = x^{12}$	(x^3, x^{12})
$x(x^4)$	$\frac{-1}{x^4} = \frac{1}{x^4} = x^{11}$	(x^4, x^{11})
$x(x^5)$	$\frac{-1}{x^5} = \frac{1}{x^5} = x^{10}$	(x^5, x^{10})
$x(x^6)$	$\frac{-1}{x^6} = \frac{1}{x^6} = x^9$	(x^6, x^9)
$x(x^7)$	$\frac{-1}{x^7} = \frac{1}{x^7} = x^8$	(x^7, x^8)
$x(x^{15})$	$\frac{-1}{x^{15}} = \frac{1}{x^{15}} = x^{15}$	(x^{15})
$x(0)$	$\frac{-1}{0} = \frac{1}{0} = \infty$	$(0, \infty)$

Table 3

Transformation $\boldsymbol{y(z)}$	**Action**	**Permutation**
$y(x)$	$\frac{x+1}{x} = x^3$ $y(x^3) = \frac{x^3+1}{x^3} = x^{11}$	(x, x^3, x^{11})
$y(x^2)$	$\frac{x^2+1}{x^2} = x^6$ $y(x^6) = \frac{x^6+1}{x^6} = x^7$	(x^2, x^6, x^7)
$y(x^4)$	$\frac{x^4+1}{x^4} = x^{12}$ $y(x^{12}) = \frac{x^{12}+1}{x^{12}} = x^{14}$	(x^4, x^{12}, x^{14})
$y(x^5)$	$\frac{x^5+1}{x^5} = x^5$ $y(x^5) = x^5$	(x^5)

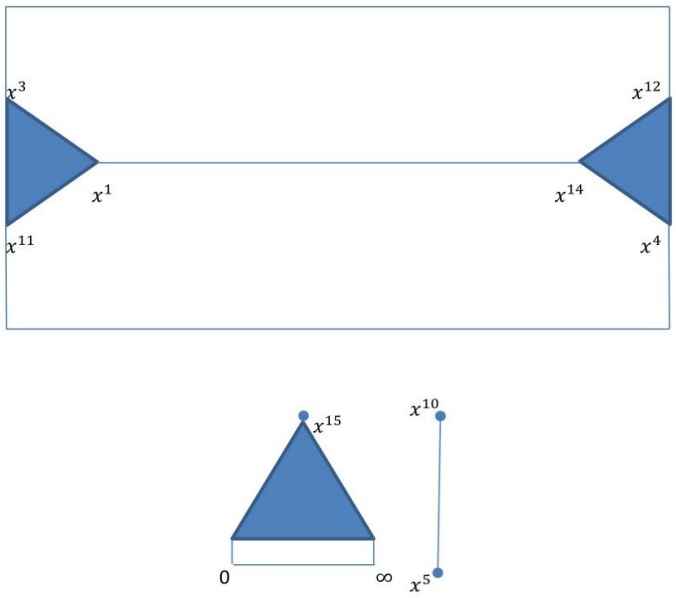
$y(x^{10})$	$\frac{x^{10}+1}{x^{10}} = x^{10}$ $y(x^{10}) = x^{10}$	(x^{10})
$y(x^{8})$	$\frac{x^{8}+1}{x^{8}} = x^{9}$ $y(x^{9}) = \frac{(x^{9}+1)}{x^{9}} = x^{13}$	(x^{8}, x^{9}, x^{13})
$y(x^{15})$	$y(x^{15}) = \frac{(x^{15}+1)}{x^{15}} = 0$ $y(0) = \frac{(0+1)}{0} = \infty$	$(x^{15}, 0, \infty)$

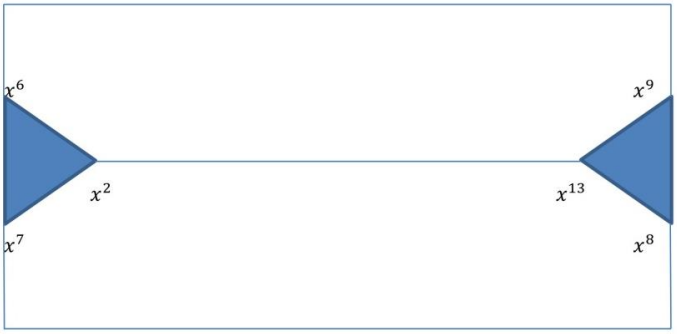
Using Table 2, 3 we have the following permutations x and y:

$$x = (x, x^{14})(x^{2}, x^{13})(x^{3}, x^{12})(x^{4}, x^{11})(x^{5}, x^{10})(x^{6}, x^{9})(x^{7}, x^{8})(x^{15})(0, \infty)$$

$$y = (x, x^{3}, x^{11})(x^{2}, x^{6}, x^{7})(x^{4}, x^{12}, x^{14})(x^{5})(x^{10})(x^{8}, x^{9}, x^{13})(x^{15}, 0, \infty)$$

These permutations combine to produce the following coset graph:





2. *Secound Galois Field F_2 of Order 16*

The Table 4 displays all elements of $GF\ (2^4)$ evolved through $f(x) = 1 + x^3 + x^4$.

Table 4

Elements	**Polynomial Description**	**Digital Description**	**Elements**	**Description**	**Digital Description**
x	x	0100	x^9	$x^2 + 1$	1010
x^2	x^2	0010	x^{10}	$x + x^3$	0101
x^3	x^3	0001	x^{11}	$x^2 + x^3 + 1$	1011
x^4	$x^3 + 1$	1001	x^{12}	$1 + x$	1100
x^5	$1 + x + x^3$	1101	x^{13}	$x + x^2$	0110
x^6	$x + x^2 + x^3 + 1$	1111	x^{14}	$x^2 + x^3$	0011
x^7	$x + x^2$	1110	x^{15}	1	1000
x^8	$x + x^2 + x^3$	0111			

In Table 5, 6 we apply the generators x and y of $H(3)$ on all elements of F_2.

Table 5

Transformation $x(z)$	**Action**	**Permutation**
$x(x)$	$\frac{-1}{x} = \frac{1}{x} = x^{14}$	(x, x^{14})
$x(x^2)$	$\frac{-1}{x^2} = \frac{1}{x^2} = x^{13}$	(x^2, x^{13})
$x(x^3)$	$\frac{-1}{x^3} = \frac{1}{x^3} = x^{12}$	(x^3, x^{12})
$x(x^4)$	$\frac{-1}{x^4} = \frac{1}{x^4} = x^{11}$	(x^4, x^{11})
$x(x^5)$	$\frac{-1}{x^5} = \frac{1}{x^5} = x^{10}$	(x^5, x^{10})

$x(x^6)$	$\frac{-1}{x^6} = \frac{1}{x^6} = x^9$	(x^6, x^9)
$x(x^7)$	$\frac{-1}{x^7} = \frac{1}{x^7} = x^8$	(x^7, x^8)
$x(x^{15})$	$\frac{-1}{x^{15}} = \frac{1}{x^{15}} = x^{15}$	(x^{15})
$x(0)$	$\frac{-1}{0} = \frac{1}{0} = \infty$	$(0, \infty)$

Table 6

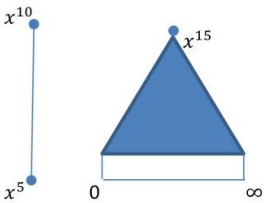
Transformation $\boldsymbol{y(z)}$	**Action**	**Permutation**
$y(x)$	$\frac{x+1}{x} = x^{11}$ $y(x^{11}) = \frac{x^{11}+1}{x^{11}} = x^3$	(x, x^{11}, x^3)
$y(x^2)$	$\frac{x^2+1}{x^2} = x^7$ $y(x^7) = \frac{x^7+1}{x^7} = x^6$	(x^2, x^7, x^6)
$y(x^4)$	$\frac{x^4+1}{x^4} = x^{14}$ $y(x^{14}) = \frac{x^{14}+1}{x^{14}} = x^{12}$	(x^4, x^{14}, x^{12})
$y(x^5)$	$\frac{x^5+1}{x^5} = x^5$ $y(x^5) = x^5$	(x^5)
$y(x^{10})$	$\frac{x^{10}+1}{x^{10}} = x^{10}$ $y(x^{10}) = x^{10}$	(x^{10})
$y(x^8)$	$\frac{x^8+1}{x^8} = x^{13}$ $y(x^{13}) = \frac{(x^{13}+1)}{x^{13}} = x^9$	$(x^8, x^{13} x^9)$
$y(x^{15})$	$y(x^{15}) = \frac{(x^{15}+1)}{x^{15}} = 0$ $y(0) = \frac{(0+1)}{0} = \infty$	$(x^{15}, 0, \infty)$

Using Table 5, 6 we have the following permutations x and y

$$x = (x, x^{14})(x^2, x^{13})(x^3, x^{12})(x^4, x^{11})(x^5, x^{10})(x^6, x^9)(x^7, x^8)(x^{15})(0, \infty)$$

$$y = (x, x^{11}, x^3)(x^2, x^7, x^6)(x^4, x^{14}, x^{12})(x^5)(x^{10})(x^8, x^{13} x^9)(x^{15}, 0, \infty)$$

These permutations combine to produce the following coset graph:



Chapter Four

Construction of $GF\ (3^3)$ and Coset graphs

In this chapter, we will construct the all Galois fields of order 27. Then we compose coset graphs for Hecke group $H(3) = \langle x, y: x^2 = y^3 = 1\rangle$ over these fields.

To construct $GF\ (3^3)$, first we need to find all irreducible polynomial of degree 3 over Z_3. All such Polynomials are given below:

i. $f(x) = 1 + 2x + x^2 + x^3$

ii. $f(x) = 1 + 2x + x^3$

iii. $f(x) = 1 + 2x^2 + x^3$

iv. $f(x) = 1 + x + 2x^2 + x^3$

1. First Galois Field F_1 of Order 27

The Table 1 displays all elements of $GF\ (3^3)$ evolved through $f(x) = 1 + 2x + x^2 + x^3$

Table 1

Elements	**Polynomial Description**	**Digital Description**	**Elements**	**Polynomial Description**	**Digital Description**
x	x	010	x^{15}	$2x^2$	002
x^2	x^2	001	x^{16}	$1 + 2x + x^2$	121
x^3	$x + 2 + 2x^2$	212	x^{17}	$x^2 + 2 + 2x$	221
x^4	$x + 1 + 2x^2$	112	x^{18}	$2 + x^2$	201
x^5	$1 + 2x^2$	102	x^{19}	$2 + 2x^2$	202
x^6	$1 + x^2$	101	x^{20}	$x + 1 + x^2$	111
x^7	$2 + 2x + 2x^2$	222	x^{21}	$2 + 2x$	220
x^8	$x + 1$	110	x^{22}	$2x + 2x^2$	022
x^9	$x + x^2$	011	x^{23}	$2x + 1$	120
x^{10}	$x + 2$	210	x^{24}	$x + 2x^2$	012
x^{11}	$2x + x^2$	021	x^{25}	$1 + 2x + 2x^2$	122
x^{12}	$x + 2 + x^2$	211	x^{26}	1	100
x^{13}	2	200			
x^{14}	$2x$	020			

In Table 2, 3 we apply the generators x and y of $H(3)$ on all elements of F_3

Table 2

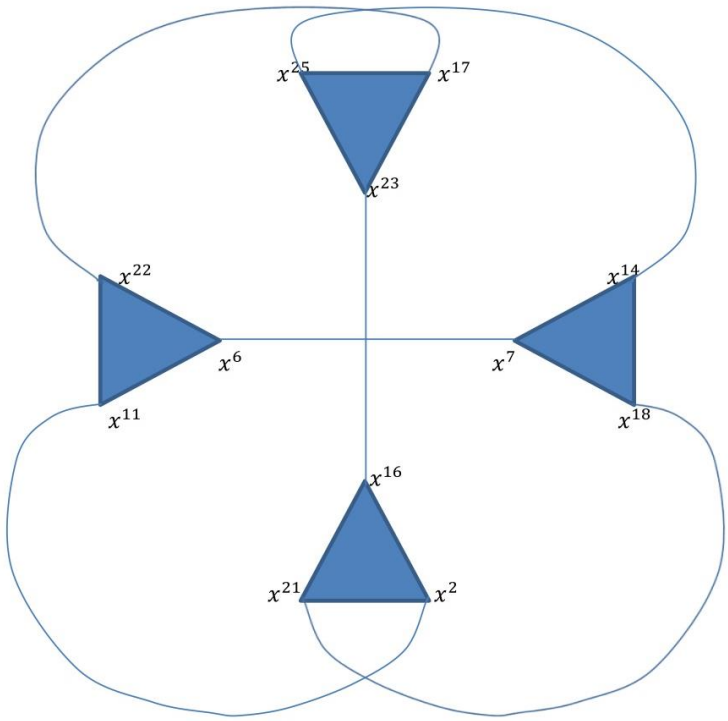
Transformation $\boldsymbol{x(z)}$	**Action**	**Permutation**
$x(x)$	$\frac{-1}{x} = \frac{2}{x} = x^{12}$	(x, x^{12})
$x(x^2)$	$\frac{-1}{x^2} = \frac{2}{x^2} = x^{11}$	(x^2, x^{11})
$x(x^3)$	$\frac{-1}{x^3} = \frac{2}{x^3} = x^{10}$	(x^3, x^{10})
$x(x^4)$	$\frac{-1}{x^4} = \frac{2}{x^4} = x^9$	(x^4, x^9)
$x(x^5)$	$\frac{-1}{x^5} = \frac{2}{x^5} = x^8$	(x^5, x^8)
$x(x^6)$	$\frac{-1}{x^6} = \frac{2}{x^6} = x^7$	(x^6, x^7)
$x(x^{13})$	$\frac{-1}{x^{13}} = \frac{2}{x^{13}} = x^{26}$	(x^{13}, x^{26})
$x(x^{14})$	$\frac{-1}{x^{14}} = \frac{2}{x^{14}} = x^{25}$	(x^{14}, x^{25})
$x(x^{15})$	$\frac{-1}{x^{15}} = \frac{2}{x^{15}} = x^{24}$	(x^{15}, x^{24})
$x(x^{16})$	$\frac{-1}{x^{16}} = \frac{2}{x^{16}} = x^{23}$	(x^{16}, x^{23})
$x(x^{17})$	$\frac{-1}{x^{17}} = \frac{2}{x^{17}} = x^{22}$	(x^{17}, x^{22})
$x(x^{18})$	$\frac{-1}{x^{18}} = \frac{2}{x^{18}} = x^{21}$	(x^{18}, x^{21})
$x(x^{19})$	$\frac{-1}{x^{19}} = \frac{2}{x^{19}} = x^{20}$	(x^{19}, x^{20})
$x(0)$	$\frac{-1}{0} = \frac{2}{0} = \infty$	$(0, \infty)$

Table 2

Transformation $\boldsymbol{y(z)}$	**Action**	**Permutation**
$y(x)$	$\frac{x+2}{x} = x^9$ $y(x^9) = \frac{x^9+2}{x^9} = x^3$	(x, x^9, x^3)
$y(x^2)$	$\frac{x^2+2}{x^2} = x^{16}$ $y(x^{16}) = \frac{x^{16}+2}{x^{16}} = x^{21}$	(x^2, x^{16}, x^{21})
$y(x^4)$	$\frac{x^4+2}{x^4} = x^{20}$ $y(x^{20}) = \frac{x^{20}+2}{x^{20}} = x^{15}$	(x^4, x^{20}, x^{15})
$y(x^5)$	$\frac{x^5+2}{x^5} = x^{10}$ $y(x^{10}) = \frac{x^{10}+2}{x^{10}} = x^{24}$	(x^5, x^{10}, x^{24})
$y(x^6)$	$\frac{x^6+2}{x^6} = x^{22}$ $y(x^{22}) = \frac{x^{22}+2}{x^{22}} = x^{11}$	(x^6, x^{22}, x^{11})
$y(x^7)$	$\frac{x^7+2}{x^7} = x^{18}$ $y(x^{18}) = \frac{x^{18}+2}{x^{18}} = x^{14}$	(x^7, x^{18}, x^{14})
$y(x^8)$	$\frac{x^8+2}{x^8} = x^{19}$ $y(x^{19}) = \frac{x^{19}+2}{x^{19}} = x^{12}$	(x^8, x^{19}, x^{12})
$y(x^{26})$	$\frac{x^{26}+2}{x^{26}} = 0$ $y(0) = \frac{0+2}{0} = \infty$	$(x^{26}, 0, \infty)$
$y(x^{17})$	$\frac{x^{17}+2}{x^{17}} = x^{25}$ $y(x^{25}) = \frac{x^{25}+2}{x^{25}} = x^{23}$	(x^{17}, x^{25}, x^{23})
$y(x^{13})$	$y(x^{13}) = \frac{x^{13}+2}{x^{13}} = x^{13}$	(x^{13})

Using Table 2, 3 we have the following permutations of x and y

$$x = (x, x^{12})(x^2, x^{11})(x^3, x^{10})(x^4, x^9)(x^5, x^8)(x^6, x^7)(x^{13}, x^{26})$$
$$(x^{14}, x^{25})(x^{15}, x^{24})(x^{16}, x^{23})(x^{17}, x^{22})(x^{18}, x^{21})(x^{19}, x^{20})(0, \infty)$$



2. Secound Galois Field F_2 of Order 27

The Table 4 displays all elements of $GF\ (3^3)$ evolved through $f(x) = 1 + 2x + x^3$

Table 4

Elements	**Polynomial Description**	**Digital Description**	**Elements**	**Polynomial Description**	**Digital Description**
x	x	010	x^{15}	$2x^2$	002
x^2	x^2	001	x^{16}	$2x + 1$	120
x^3	$x + 2$	210	x^{17}	$2x^2 + x$	012
x^4	$x^2 + 2x$	021	x^{18}	$2x + 1 + x^2$	121
x^5	$x + 2 + 2x^2$	212	x^{19}	$2x^2 + x + 2$	222
x^6	$x^2 + x + 1$	111	x^{20}	$2x^2 + x + 1$	112
x^7	$x^2 + 2x + 2$	221	x^{21}	$x^2 + 1$	101
x^8	$2x^2 + 2$	202	x^{22}	$2x + 2$	220
x^9	$x + 1$	110	x^{23}	$2x^2 + 2x$	022
x^{10}	$x^2 + x$	011	x^{24}	$2x^2 + 2x + 1$	122
x^{11}	$x^2 + x + 2$	211	x^{25}	$2x^2 + 1$	102
x^{12}	$x^2 + 2$	201	x^{26}	1	100
x^{13}	2	200			
x^{14}	$2x$	020			

we apply the generators x and y of $H(3)$ on all elements of F_2

Table 5

Transformation $\boldsymbol{x(z)}$	**Action**	**Permutation**
$x(x)$	$\frac{-1}{x} = \frac{2}{x} = x^{12}$	(x, x^{12})
$x(x^2)$	$\frac{-1}{x^2} = \frac{2}{x^2} = x^{11}$	(x^2, x^{11})
$x(x^3)$	$\frac{-1}{x^3} = \frac{2}{x^3} = x^{10}$	(x^3, x^{10})
$x(x^4)$	$\frac{-1}{x^4} = \frac{2}{x^4} = x^9$	(x^4, x^9)
$x(x^5)$	$\frac{-1}{x^5} = \frac{2}{x^5} = x^8$	(x^5, x^8)
$x(x^6)$	$\frac{-1}{x^6} = \frac{2}{x^6} = x^7$	(x^6, x^7)
$x(x^{13})$	$\frac{-1}{x^{13}} = \frac{2}{x^{13}} = x^{26}$	(x^{13}, x^{26})
$x(x^{14})$	$\frac{-1}{x^{14}} = \frac{2}{x^{14}} = x^{25}$	(x^{14}, x^{25})
$x(x^{15})$	$\frac{-1}{x^{15}} = \frac{2}{x^{15}} = x^{24}$	(x^{15}, x^{24})
$x(x^{16})$	$\frac{-1}{x^{16}} = \frac{2}{x^{16}} = x^{23}$	(x^{16}, x^{23})
$x(x^{17})$	$\frac{-1}{x^{17}} = \frac{2}{x^{17}} = x^{22}$	(x^{17}, x^{22})
$x(x^{18})$	$\frac{-1}{x^{18}} = \frac{2}{x^{18}} = x^{21}$	(x^{18}, x^{21})
$x(x^{19})$	$\frac{-1}{x^{19}} = \frac{2}{x^{19}} = x^{20}$	(x^{19}, x^{20})
$x(0)$	$\frac{-1}{0} = \frac{2}{0} = \infty$	$(0, \infty)$

Table 6

Transformation $\boldsymbol{y(z)}$	**Action**	**Permutation**
$y(x)$	$\frac{x+2}{x} = x^2$ $y(x^2) = \frac{x^2+2}{x^2} = x^{10}$	$(x,,x^2,x^{10})$
$y(x^3)$	$\frac{x^3+2}{x^3} = x^6$ $y(x^6) = \frac{x^6+2}{x^6} = x^4$	(x^3,x^6,x^4)
$y(x^5)$	$\frac{x^5+2}{x^5} = x^{15}$ $y(x^{15}) = \frac{x^{15}+2}{x^{15}} = x^{19}$	(x^5,x^{15},x^{19})
$y(x^7)$	$\frac{x^7+2}{x^7} = x^{11}$ $y(x^{11}) = \frac{x^{11}+2}{x^{11}} = x^{21}$	(x^7,x^{11},x^{21})
$y(x^8)$	$\frac{x^8+2}{x^8} = x^{17}$ $y(x^{17}) = \frac{x^{17}+2}{x^{17}} = x^{14}$	(x^8,x^{17},x^{14})
$y(x^9)$	$\frac{x^9+2}{x^9} = x^{21}$ $y(x^{21}) = \frac{x^{21}+2}{x^{21}} = x^{12}$	(x^9,x^{21},x^{12})
$y(x^{16})$	$\frac{x^{16}+2}{x^{16}} = x^{24}$ $y(x^{24}) = \frac{x^{24}+2}{x^{24}} = x^{25}$	(x^{16},x^{24},x^{25})
$y(x^{13})$	$\frac{x^{13}+2}{x^{13}} = x^{13}$ $y(x^{13}) = \frac{x^{13}+2}{x^{13}} = x^{13}$	(x^{13})
$y(x^{20})$	$\frac{x^{20}+2}{x^{20}} = x^{23}$ $y(x^{23}) = \frac{x^{23}+2}{x^{23}} = x^{22}$	(x^{20},x^{23},x^{22})
$y(x^{26})$	$\frac{x^{26}+2}{x^{26}} = 0$ $y(0) = \frac{0+2}{0} = \infty$	$(x^{26},0,\infty)$

By using Table 5, 6 we have the following permutations of x and y

$$x = (x, x^{12})(x^2, x^{11})(x^3, x^{10})(x^4, x^9)(x^5, x^8)(x^6, x^7)(x^{13}, x^{26})$$
$$(x^{14}, x^{25})(x^{15}, x^{24})(x^{16}, x^{23})(x^{17}, x^{22})(x^{18}, x^{21})(x^{19}, x^{20})(0, \infty)$$

$$y = (x,, x^2, x^{10})(x^3, x^6, x^4)(x^5, x^{15}, x^{19})(x^7, x^{11}, x^{21})(x^8, x^{17}, x^{14})(x^9, x^{21}, x^{12})$$
$$(x^{16}, x^{24}, x^{25})(x^{13})(x^{20}, x^{23}, x^{22})\ (x^{26}, 0, \infty)$$

Now by using the permutation of x and y one can easily make corresponding coset graphs

3. *Third Galois Field F_3 of Order 27*

The Table 7 displays all elements of $GF\ (3^3)$ evolved through $f(x) = 1 + 2x^2 + x^3$

Table 7

Elements	**Polynomial Description**	**Digital Description**	**Elements**	**Polynomial Description**	**Digital Description**
x	x	010	x^{15}	$2x^2$	002
x^2	x^2	001	x^{16}	$1 + 2x^2$	102
x^3	$2 + x^2$	201	x^{17}	$x + 1 + 2x^2$	112
x^4	$2 + 2x + 2x^2$	221	x^{18}	$1 + x$	110
x^5	$2 + 2x$	220	x^{19}	$x + x^2$	011
x^6	$2x + 2x^2$	022	x^{20}	$2 + 2x^2$	202
x^7	$1 + x^2$	101	x^{21}	$1 + 2x + 2x^2$	122
x^8	$2 + x + x^2$	211	x^{22}	$x + 1 + x^2$	111
x^9	$2 + 2x + 2x^2$	222	x^{23}	$2 + x + 2x^2$	212
x^{10}	$1 + 2x + x^2$	121	x^{24}	$1 + 2x$	120
x^{11}	$x + 2$	210	x^{25}	$x + 2x^2$	012
x^{12}	$2x + x^2$	021	x^{26}	1	100
x^{13}	2	200			
x^{14}	$2x$	020			

we apply the generators x and y of $H(3)$ on all elements of F_3

Table 8

Transformation $x(z)$	**Action**	**Permutation**
$x(x)$	$\frac{-1}{x} = \frac{2}{x}$	(x, x^{12})
$x(x^2)$	$\frac{-1}{x^2} = \frac{2}{x^2}$	(x^2, x^{11})
$x(x^3)$	$\frac{-1}{x^3} = \frac{2}{x^3}$	(x^3, x^{10})

$x(x^4)$	$\frac{-1}{x^4} = \frac{2}{x^4}$	(x^4, x^9)
$x(x^5)$	$\frac{-1}{x^5} = \frac{2}{x^5}$	(x^5, x^8)
$x(x^6)$	$\frac{-1}{x^6} = \frac{2}{x^6}$	(x^6, x^7)
$x(x^{13})$	$\frac{-1}{x^{13}} = \frac{2}{x^{13}}$	(x^{13}, x^{26})
$x(x^{14})$	$\frac{-1}{x^{14}} = \frac{2}{x^{14}}$	(x^{14}, x^{25})
$x(x^{15})$	$\frac{-1}{x^{15}} = \frac{2}{x^{15}}$	(x^{15}, x^{24})
$x(x^{16})$	$\frac{-1}{x^{16}} = \frac{2}{x^{16}}$	(x^{16}, x^{23})
$x(x^{17})$	$\frac{-1}{x^{17}} = \frac{2}{x^{17}}$	(x^{17}, x^{22})
$x(x^{18})$	$\frac{-1}{x^{18}} = \frac{2}{x^{18}}$	(x^{18}, x^{21})
$x(x^{19})$	$\frac{-1}{x^{19}} = \frac{2}{x^{19}}$	(x^{19}, x^{20})
$x(0)$	$\frac{-1}{0} = \frac{2}{0} = \infty$	$(0, \infty)$

Table 9

Transformation $\boldsymbol{y(z)}$	**Action**	**Permutation**
$y(x)$	$\frac{x+2}{x} = x^{10}$ $y(x^{10}) = \frac{x^{10}+2}{x^{10}} = x^2$	(x, x^{10}, x^2)
$y(x^3)$	$\frac{x^3+2}{x^3} = x^4$ $y(x^4) = \frac{x^4+2}{x^4} = x^6$	(x^3, x^4, x^6)
$y(x^5)$	$\frac{x^5+2}{x^5} = x^{19}$	(x^5, x^{19}, x^{15})

	$y(x^{19}) = \frac{x^{19}+2}{x^{19}} = x^{15}$	
$y(x^7)$	$\frac{x^7+2}{x^7} = x^{21}$ $y(x^{21}) = \frac{x^{21}+2}{x^{21}} = x^{11}$	(x^7, x^{21}, x^{11})
$y(x^8)$	$\frac{x^8+2}{x^8} = x^{14}$ $y(x^{14}) = \frac{x^{14}+2}{x^{14}} = x^{17}$	(x^8, x^{14}, x^{17})
$y(x^9)$	$\frac{x^9+2}{x^9} = x^{12}$ $y(x^{12}) = \frac{x^{12}+2}{x^{12}} = x^{21}$	(x^9, x^{12}, x^{21})
$y(x^{16})$	$\frac{x^{16}+2}{x^{16}} = x^{25}$ $y(x^{25}) = \frac{x^{25}+2}{x^{25}} = x^{24}$	(x^{16}, x^{25}, x^{24})
$y(x^{13})$	$\frac{x^{13}+2}{x^{13}} = x^{13}$ $y(x^{13}) = \frac{x^{13}+2}{x^{13}} = x^{13}$	(x^{13})
$y(x^{20})$	$\frac{x^{20}+2}{x^{20}} = x^{22}$ $y(x^{22}) = \frac{x^{22}+2}{x^{22}} = x^{23}$	(x^{20}, x^{22}, x^{23})
$y(x^{26})$	$\frac{x^{26}+2}{x^{26}} = 0$ $y(0) = \frac{0+2}{0} = \infty$	$(x^{26}, 0, \infty)$

By using table 8, 9 we have following permutations of x and y

$$x = (x, x^{12})(x^2, x^{11})(x^3, x^{10})(x^4, x^9)(x^5, x^8)(x^6, x^7)(x^{13}, x^{26})$$
$$(x^{14}, x^{25})(x^{15}, x^{24})(x^{16}, x^{23})(x^{17}, x^{22})(x^{18}, x^{21})(x^{19}, x^{20})(0, \infty)$$

$$y = (x, x^{10}, x^2)(x^3, x^4, x^6)(x^5, x^{19}, x^{15})(x^7, x^{21}, x^{11})(x^8, x^{14}, x^{17})$$
$$(x^9, x^{12}, x^{21})(x^{16}, x^{25}, x^{24})(x^{13})(x^{20}, x^{22}, x^{23})(x^{26}, 0, \infty)$$

Now, by using the permutation of x and y one can easily make corresponding coset graphs

4 Fourth Galois Field F_4 of Order 27

The Table 10 displays all elements of $GF\ (3^3)$ evolved through
$f(x) = 1 + x + 2x^2 + x^3$

Table 10

Elements	**Polynomial Description**	**Digital Description**	**Elements**	**Polynomial Description**	**Digital Description**
x	x	010	x^{15}	$2x^2$	002
x^2	x^2	001	x^{16}	$x+1+2x^2$	112
x^3	x^2+2+2x	221	x^{17}	$2x+1$	120
x^4	$2+x$	210	x^{18}	$x+2x^2$	012
x^5	x^2+2x	021	x^{19}	$1+x$	110
x^6	$2x+2$	220	x^{20}	$x+x^2$	011
x^7	$2x^2+2x$	022	x^{21}	$2+2x+2x^2$	222
x^8	$1+x+x^2$	111	x^{22}	$1+x^2$	101
x^9	$2+2x^2$	202	x^{23}	$2+x^2$	201
x^{10}	$1+2x^2$	102	x^{24}	$2+x+x^2$	211
x^{11}	$2x^2+2x+1$	122	x^{25}	$2+x+2x^2$	212
x^{12}	x^2+2x+1	121	x^{26}	1	100
x^{13}	2	200			
x^{14}	$2x$	020			

we apply the generators x and y of $H(3)$ on all elements of F_4

Table 11

Transformation $\boldsymbol{x(z)}$	**Action**	**Permutation**
$x(x)$	$\frac{-1}{x}=\frac{2}{x}=x^{12}$	(x,x^{12})
$x(x^2)$	$\frac{-1}{x^2}=\frac{2}{x^2}=x^{11}$	(x^2,x^{11})
$x(x^3)$	$\frac{-1}{x^3}=\frac{2}{x^3}=x^{10}$	(x^3,x^{10})
$x(x^4)$	$\frac{-1}{x^4}=\frac{2}{x^4}=x^{9}$	(x^4,x^{9})
$x(x^5)$	$\frac{-1}{x^5}=\frac{2}{x^5}=x^{8}$	(x^5,x^{8})
$x(x^6)$	$\frac{-1}{x^6}=\frac{2}{x^6}=x^{7}$	(x^6,x^{7})
$x(x^{13})$	$\frac{-1}{x^{13}}=\frac{2}{x^{13}}=x^{26}$	(x^{13},x^{26})

$x(x^{14})$	$\frac{-1}{x^{14}} = \frac{2}{x^{14}} = x^{25}$	(x^{14}, x^{25})
$x(x^{15})$	$\frac{-1}{x^{15}} = \frac{2}{x^{15}} = x^{24}$	(x^{15}, x^{24})
$x(x^{16})$	$\frac{-1}{x^{16}} = \frac{2}{x^{16}} = x^{23}$	(x^{16}, x^{23})
$x(x^{17})$	$\frac{-1}{x^{17}} = \frac{2}{x^{17}} = x^{22}$	(x^{17}, x^{22})
$x(x^{18})$	$\frac{-1}{x^{18}} = \frac{2}{x^{18}} = x^{21}$	(x^{18}, x^{21})
$x(x^{19})$	$\frac{-1}{x^{19}} = \frac{2}{x^{19}} = x^{20}$	(x^{19}, x^{20})
$x(0)$	$\frac{-1}{0} = \frac{2}{0} = \infty$	$(0, \infty)$

Table 12

Transformation $\boldsymbol{y(z)}$	**Action**	**Permutation**
$y(x)$	$\frac{x+2}{x} = x^3$ $y(x^3) = \frac{x^3+2}{x^3} = x^9$	(x, x^3, x^9)
$y(x^2)$	$\frac{x^2+2}{x^2} = x^{21}$ $y(x^{21}) = \frac{x^{21}+2}{x^{21}} = x^{16}$	(x^2, x^{21}, x^{16})
$y(x^4)$	$\frac{x^4+2}{x^4} = x^{15}$ $y(x^{15}) = \frac{x^{15}+2}{x^{15}} = x^{20}$	(x^4, x^{15}, x^{20})
$y(x^5)$	$\frac{x^5+2}{x^5} = x^{24}$ $y(x^{24}) = \frac{x^{24}+2}{x^{24}} = x^{10}$	(x^5, x^{24}, x^{10})
$y(x^6)$	$\frac{x^6+2}{x^6} = x^{11}$ $y(x^{11}) = \frac{x^{11}+2}{x^{11}} = x^{22}$	(x^6, x^{11}, x^{22})
$y(x^7)$	$\frac{x^7+2}{x^7} = x^{14}$	(x^7, x^{14}, x^{18})

	$y(x^{14}) = \frac{x^{14}+2}{x^{14}} = x^{18}$	
$y(x^8)$	$\frac{x^8+2}{x^8} = x^{12}$ $y(x^{12}) = \frac{x^{12}+2}{x^{12}} = x^{19}$	(x^8, x^{12}, x^{19})
$y(x^{13})$	$\frac{x^{13}+2}{x^{13}} = x^{13}$ $y(x^{13}) = \frac{x^{13}+2}{x^{13}} = x^{13}$	(x^{13})
$y(x^{17})$	$\frac{x^{17}+2}{x^{17}} = x^{23}$ $y(x^{23}) = \frac{x^{23}+2}{x^{23}} = x^{25}$	(x^{17}, x^{23}, x^{25})
$y(x^{26})$	$\frac{x^{26}+2}{x^{26}} = 0$ $y(0) = \frac{0+2}{0} = \infty$	$(x^{26}, 0, \infty)$

By using table 11, 12 we can make following permutations of x and y

$$x = (x, x^{12})(x^2, x^{11})(x^3, x^{10})(x^4, x^9)(x^5, x^8)(x^6, x^7)(x^{13}, x^{26})$$
$$(x^{14}, x^{25})(x^{15}, x^{24})(x^{16}, x^{23})(x^{17}, x^{22})(x^{18}, x^{21})(x^{19}, x^{20})(0, \infty)$$

$$y = (x, x^3, x^9)(x^2, x^{21}, x^{16})(x^4, x^{15}, x^{20})(x^5, x^{24}, x^{10})(x^6, x^{11}, x^{22})$$
$$(x^7, x^{14}, x^{18})(x^8, x^{12}, x^{19})(x^{13})(x^{17}, x^{23}, x^{25})(x^{26}, 0, \infty)$$

Now, by using the permutation of x and y one can easily make corresponding coset graphs

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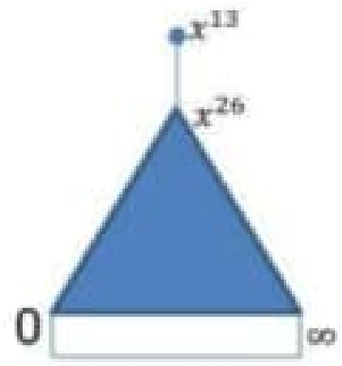
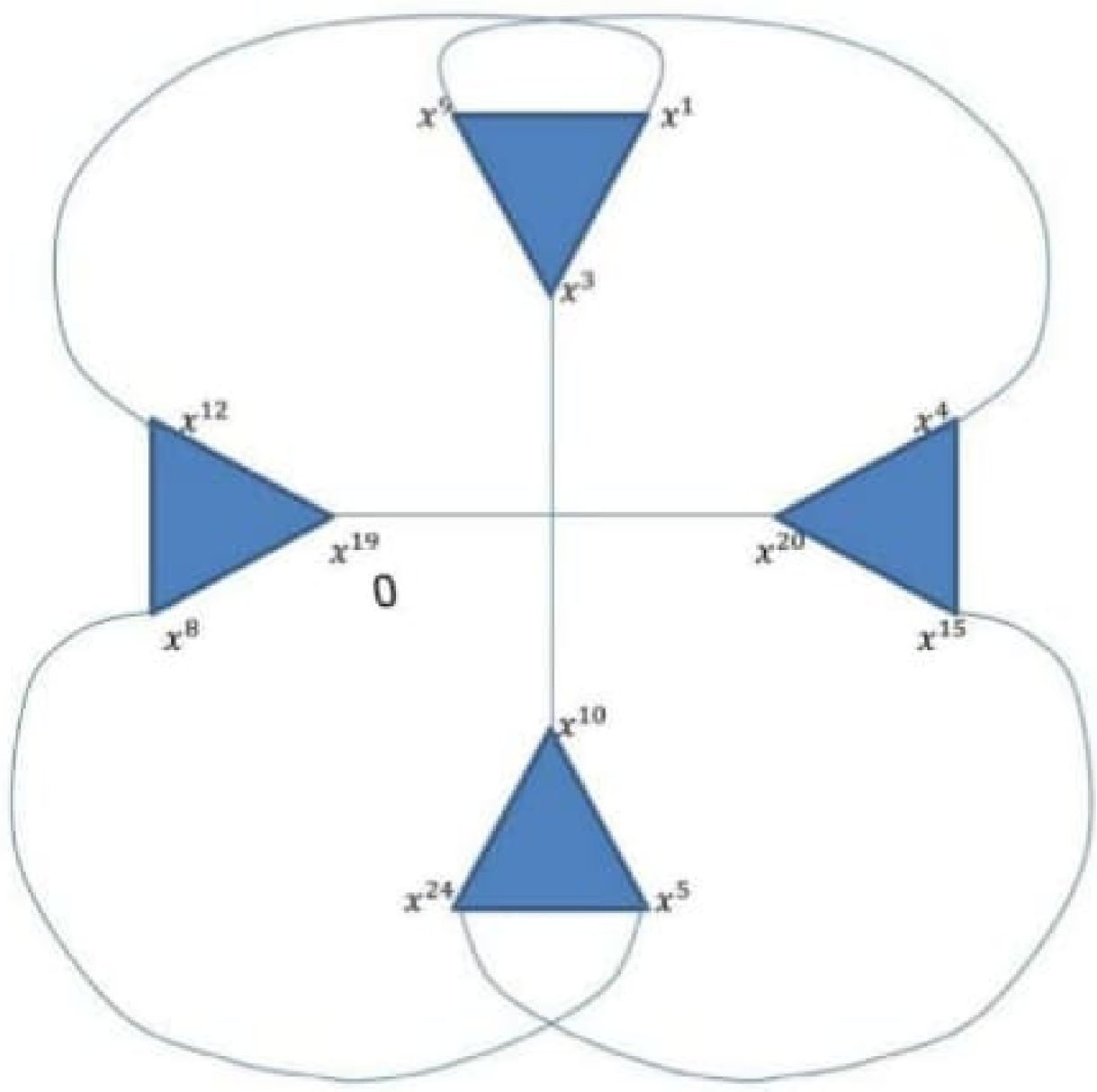
$$y = (x, x^9, x^3)(x^2, x^{16}, x^{21})(x^4, x^{20}, x^{15})(x^5, x^{10}, x^{24})(x^6, x^{22}, x^{11})$$
$$(x^7, x^{18}, x^{14})(x^8, x^{19}, x^{12})(x^{26}, 0, \infty)(x^{17}, x^{25}, x^{23})(x^{13})$$

These permutations combine to produce the following Coset graphs





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